

Shang Liu

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EDUCATION

Ph.D. in Operations and Analytics

Business School, Imperial College London (ICL), London, UK.

Sept. 2022 – Jun. 2027 (Expected)

Advisors: Prof. Xiaocheng Li and Prof. Kalyan Talluri

M.Res. in Operations and Analytics (Distinction)

Business School, Imperial College London (ICL), London, UK.

Sept. 2021 – Jun. 2022

B.S. in Mathematics and Applied Mathematics

School of Mathematical Sciences, Peking University (PKU), Beijing, China.

Sept. 2016 – Jun. 2021

RESEARCH INTERESTS

My research focuses on enhancing the alignment, robustness, and ethical deployment of large language models (LLMs) through interdisciplinary frameworks of human-AI collaboration, game theory, statistical learning, and optimization. I aim to develop both theoretically guaranteed (e.g., convergence analyses, complexity bounds) and empirically valid methodologies. My research interests also lie in the development and theoretical analysis of machine learning models, with a particular focus on the role of uncertainty, including uncertainty quantification, active learning, and human-AI interaction.

PUBLICATIONS

(reverse chronological order; *: equal contribution)

Wasserstein Distributionally Robust Regret Optimization for Reinforcement Learning from Human Feedback

Yikai Wang*, Shang Liu*, Jose Blanchet, [arXiv](#)

Incentivizing High-Quality Human Annotations with Golden Questions

Shang Liu, Zhongze Cai, Hanzhao Wang, Zhongyao Ma, Xiaocheng Li, [arXiv](#)

How Humans Help LLMs: Assessing and Incentivizing Human Preference Annotators

Shang Liu*, Hanzhao Wang*, Zhongyao Ma, Xiaocheng Li, *under Minor Revision at Management Science*, [arXiv](#)

Reward Modeling with Ordinal Feedback: Wisdom of the Crowd

Shang Liu*, Yu Pan*, Guanting Chen, Xiaocheng Li, *ICML 2025*, [arXiv](#)

Towards Better Understanding of In-Context Learning Ability from In-Context Uncertainty Quantification
Shang Liu*, Zhongze Cai*, Guanting Chen, Xiaocheng Li, *Transactions on Machine Learning Research*, [arXiv](#)

Towards Better Statistical Understanding of Watermarking LLMs
Zhongze Cai*, Shang Liu*, Hanzhao Wang*, Huaiyang Zhong, Xiaocheng Li, *accepted at Journal of the American Statistical Association*, [arXiv](#)

When No-Rejection Learning is Consistent for Regression with Rejection
Xiaocheng Li, Shang Liu, Chunlin Sun, Hanzhao Wang, *AISTATS 2024*, [arXiv](#)

Understanding Uncertainty Sampling via Equivalent Loss
Shang Liu, Xiaocheng Li, [arXiv](#)

Distribution-Free Model-Agnostic Regression Calibration via Nonparametric Methods
Shang Liu*, Zhongze Cai*, Xiaocheng Li, *NeurIPS 2023*, [arXiv](#)

Maximum Optimality Margin: A Unified Approach for Contextual Linear Programming and Inverse Linear Programming
Chunlin Sun*, Shang Liu*, Xiaocheng Li, *ICML 2023*, [arXiv](#)

Non-stationary Bandits with Knapsacks
Shang Liu, Jiashuo Jiang, Xiaocheng Li, *NeurIPS 2022*, [arXiv](#)

Online Bin Packing with Known T
Shang Liu, Xiaocheng Li, *under Major Revision at Mathematics of Operations Research*, [arXiv](#)

PRIZES & AWARDS

Dean's List for Academic Excellence at Imperial College Business School, 2021/2022
Outstanding Graduate of Peking University, 2021
Wusi Scholarships at Peking University, 2016/2017, 2017/2018

SEVICE

Reviewer: NeurIPS, ICLR, AISTATS, ICML, TMLR, INFORMS IJDS

TEACHING

TA:
Machine Learning, Imperial College London, Spring 2024/2025/2026
Generative AI and Large Language Models, Imperial College London, Spring 2025/2026